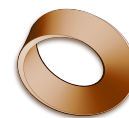


Name: _____
 Class: _____
 Date: _____



Mr. Rawson • GCSE Physics

Kinematics 1: Scalars and Vectors

The partner sheet to *Mapping Your Journey*. You will need the four numbers you measured.

1 Start with the drive

My Maps told you two things about the drive from home to school.

From My Maps	Your value	Unit
How far you drive		km
How long it takes		min

Neither needed a **direction** to make sense. “16 km” is a complete answer. So is “19 minutes”.

2 Then you drew the straight line

Google Earth gave you two more.

From Google Earth	Your value	Unit
Length of the straight line		km
Heading		°

The length alone was *not* enough — “13.6 km” does not tell anyone where you live. You had to add the direction.

And the straight line ignored **every road you drove on**. It cared only about **where you started**, **where you finished**, and the **direction** between them. That is **displacement**.

3 Scalars and vectors

Scalar: A quantity with **size only**.

Vector: A quantity with **size and direction**.

Scalars

distance
 speed
 mass
 time

Vectors

displacement
 velocity
 weight
 force

1) **Circle S** for scalar or **V** for vector.

5 m S / V

5 m north S / V

10 kg S / V

10 N down S / V

4 Position, distance and displacement

Position: Where something is.

Distance: How far it has travelled. A **scalar**.

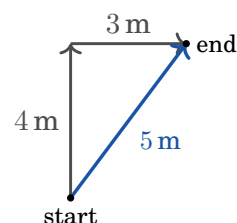
Displacement: How far it has ended up from its starting position, and in which direction. A **vector**.

Example. You walk 4 m north, then 3 m east.

Distance = $4 + 3 = 7$ m

Displacement = 5 m, north-east

You travelled 7 m, but you only ended up 5 m from where you began.



2) **State** your displacement if you walk to school and then straight home again.

5 Speed and velocity

Speed: $\text{speed} = \frac{\text{distance}}{\text{time}}$ — a **scalar**.

Velocity: $\text{velocity} = \frac{\text{displacement}}{\text{time}}$ — a **vector**.

Use your own four numbers from page 1. Give both answers in km/h.

3) **Calculate** your average **speed** on the drive to school.

4) **Calculate** your average **velocity** on the drive to school. Remember it needs a direction.

5) **Explain** why your average velocity is *smaller* than your average speed.

Mr Rawson drives 16 km in 19 minutes, and his displacement is 13.6 km north-west. His average speed is 50.5 km/h; his average velocity is 42.9 km/h north-west.